Bounding the OLS Slope of a Time Series When You Only Have Bucket Summaries

1 The problem, concretely

We have a time series of raw points (x_{ij}, y_{ij}). We want one number: the ordinary least-squares (OLS) slope of a contiguous stretch of that series, the segment. Written out, that slope is

$$\beta \;=\; \frac{S_{xy}}{S_{xx}}, \qquad S_{xx} = \sum_{i,j}(x_{ij} - \bar{X})^2, \qquad S_{xy} = \sum_{i,j}(x_{ij} - \bar{X})(y_{ij} - \bar{Y}),$$

where $\bar{X}, \bar{Y}$ are the means of all the points in the segment.

What we actually have. We do *not* have the raw points. The store has already chopped the segment into n consecutive *buckets* and thrown the raw points away, keeping for each bucket i only a five-number summary:

- its time span $[a_i, b_i]$ (exact calendar boundaries),
- how many points it holds, c_i,
- the smallest and largest y in it, ℓ_i and h_i,
- the exact mean of its y values, μ_i.

That is everything. We never see an individual x_{ij} or y_{ij} again.

What we want to produce. Because the raw points are gone, we cannot compute β exactly – many different point clouds are consistent with the same bucket summaries, and they have different slopes. So instead of a single number we will produce an **interval** $[\beta^-, \beta^+]$ that is *guaranteed* to contain the true slope, whatever the (unknown) raw points were, as long as they are consistent with the summaries. The whole proof is the construction of that interval and the argument that it really does trap β.

The one source of uncertainty. For a single bucket, the summary pins down a lot but not everything. The y side is well controlled: every y_{ij} lies in $[\ell_i, h_i]$, and we know the bucket's exact y-mean μ_i. The x side is the problem. We know every x_{ij} lies in the time span $[a_i, b_i]$, but we have *no* information about where inside that span the points fall. They could be bunched at the left edge, bunched at the right, spread out – the summary does not say. This single unknown – the horizontal placement of points inside each bucket – is the entire source of slack in everything that follows. Keep an eye on it.

2 Setting up the bookkeeping

Fix notation per bucket i:

$$m_i = \frac{a_i + b_i}{2} \quad \text{(the midpoint)}, \qquad w_i = b_i - a_i \quad \text{(the width)}, \qquad N = \sum_i c_i \quad \text{(total points)}.$$

The true (but unknown) average x-position of the points in bucket i is

$$\bar{x}_i \;=\; \frac{1}{c_i} \sum_j x_{ij}.$$

We do not know $\bar{x}_i$. All the summary tells us is that it lies somewhere in the bucket's span $[a_i, b_i]$.

Why the midpoint — a minimax choice. To compute anything we must pick a single representative x-position for bucket i to stand in for the unknown $\bar{x}_i$. Call our choice $p_i \in [a_i, b_i]$. Whatever we pick, the error we will carry is $\bar{x}_i - p_i$, and an adversary – choosing the worst point cloud consistent with the summary – may place the true $\bar{x}_i$ anywhere in $[a_i, b_i]$. So the worst-case size of our error is

$$\max_{\bar{x}_i \in [a_i, b_i]} \left| \bar{x}_i - p_i \right| = \max\big(p_i - a_i,\; b_i - p_i\big).$$

We get to pick p_i *before* the adversary picks $\bar{x}_i$, so we should pick the p_i that makes this worst case as small as possible. The maximum of the two distances $p_i - a_i$ and $b_i - p_i$ is smallest exactly when they are equal,

$$p_i - a_i = b_i - p_i \quad \iff \quad p_i = \frac{a_i + b_i}{2} = m_i,$$

i.e. the **midpoint**. Any other point in the bucket makes one of the two distances longer, and the adversary simply places $\bar{x}_i$ at that far end, giving a strictly larger guaranteed error. The midpoint is therefore not a convenience – it is the unique choice that minimizes the worst-case x-error, and with it that worst case is exactly half the bucket width:

$$\delta_i \;:=\; \bar{x}_i - m_i, \qquad |\delta_i| \le \frac{w_i}{2}. \tag{$\star$}$$

Inequality ($\star$) is the *only* thing we know about the x-placement, and the whole bound is built by pushing ($\star$) through the algebra. Because the midpoint achieves the smallest possible constant ($w_i/2$) on the right of ($\star$), every error term that later inherits this constant – E_P, $E_\downarrow$, $E_\uparrow$ – is as tight as the available information allows. Pick any other representative and all of them grow.

The y side needs no such choice: the summary stores the *exact* bucket mean μ_i, so the bucket's y-centroid is exactly μ_i (write $\bar{y}_i = \mu_i$) – an equality, not a bound.

Quantities we can actually compute. We cannot compute $\bar{X}$ (it needs the unknown $\bar{x}_i$). But with the midpoint as each bucket's x-representative, and the exact mean as each bucket's y-representative, we can compute the count-weighted surrogates

$$\hat{X} = \frac{1}{N} \sum_i c_i\, m_i, \qquad \hat{Y} = \frac{1}{N} \sum_i c_i\, \mu_i.$$

These are our stand-ins for the unknown true means $\bar{X}, \bar{Y}$. (A second, technical reason the midpoint is convenient: it makes the sums $\sum_i c_i(\mu_i - \hat{Y})$ and $\sum_i c_i u_i$ collapse to zero, which is what kills the first-order error terms in Lemmas B and C. That cancellation actually works for any affine choice of representative; the minimax argument above is the reason the midpoint in particular is the *right* affine choice.) The first job is to measure how wrong the stand-ins are.

3 Step 1 — the means (Lemma A)

Lemma 1. *$\bar{Y} = \hat{Y}$ exactly. And the x-mean error, $\Delta := \bar{X} - \hat{X}$, satisfies*

$$\Delta = \frac{1}{N}\sum_i c_i\,\delta_i, \qquad |\Delta| \le \frac{1}{N}\sum_i c_i\,\frac{w_i}{2} \;=:\; D.$$

Proof. Group the global y-average by bucket. The points in bucket i contribute total y equal to $c_i\bar{y}_i = c_i\mu_i$, so

$$\bar{Y} = \frac{1}{N}\sum_{i,j} y_{ij} = \frac{1}{N}\sum_i c_i\bar{y}_i = \frac{1}{N}\sum_i c_i\mu_i = \hat{Y}.$$

This is exact – nothing was approximated, because μ_i is the exact bucket mean. The y side carries no slack.

For x, do the same grouping but substitute $\bar{x}_i = m_i + \delta_i$ (the definition of δ_i):

$$\bar{X} = \frac{1}{N}\sum_i c_i\bar{x}_i = \frac{1}{N}\sum_i c_i\,(m_i + \delta_i) = \underbrace{\frac{1}{N}\sum_i c_i m_i}_{=\hat{X}} + \frac{1}{N}\sum_i c_i\delta_i.$$

So $\bar{X} - \hat{X} = \frac{1}{N}\sum_i c_i\delta_i$, which is the stated Δ. Bounding it: by the triangle inequality and $(\star)$,

$$|\Delta| \le \frac{1}{N}\sum_i c_i\,|\delta_i| \le \frac{1}{N}\sum_i c_i\,\frac{w_i}{2} = D. \qquad \square$$

Plain meaning: our y stand-in is perfect; our x stand-in is off by an unknown amount Δ, but that amount is provably no larger than D, a number we can compute from widths and counts alone.

4 Step 2 — separating "between" and "within"

To bound S_{xx} and S_{xy} we split each into two parts: variation *between* the buckets (driven by where the bucket centroids sit) and variation *within* each bucket (driven by the spread of points around their own centroid). This split is exact.

For any bucket the points satisfy two zero-sum identities by definition of the centroid:

$$\sum_j (x_{ij} - \bar{x}_i) = 0, \qquad \sum_j (y_{ij} - \bar{y}_i) = 0.$$

Write $x_{ij} - \bar{X} = (\bar{x}_i - \bar{X}) + (x_{ij} - \bar{x}_i)$ and similarly for y, then expand the squares/products in S_{xx}, S_{xy}. Every cross term contains a factor $\sum_j (x_{ij} - \bar{x}_i)$ or $\sum_j (y_{ij} - \bar{y}_i)$, which is zero, so the cross terms vanish and we are left with exactly:

$$S_{xx} = \underbrace{\sum_i c_i(\bar{x}_i - \bar{X})^2}_{\text{between}} + \underbrace{\sum_i A_i}_{\text{within}}, \qquad A_i = \sum_j (x_{ij} - \bar{x}_i)^2,$$

$$S_{xy} = \underbrace{\sum_i c_i(\bar{x}_i - \bar{X})(\mu_i - \bar{Y})}_{\text{between}} + \underbrace{\sum_i B_i}_{\text{within}}, \qquad B_i = \sum_j (x_{ij} - \bar{x}_i)(y_{ij} - \bar{y}_i).$$

Both identities are exact. Now we bound the two parts separately: the between part using Lemma 1 (it involves the unknown $\bar{X}$, which we replace by $\hat{X}$ at the cost of Δ), and the within part using the bucket's min/max and width.

5 Step 3 — bounding S_{xy} (Lemma B)

Define the quantity we *can* compute – the between part with the unknown true means replaced by the computable stand-ins:

$$P_0 = \sum_i c_i\,(m_i - \hat{X})(\mu_i - \hat{Y}).$$

Lemma 2. *$\left| S_{xy} - P_0 \right| \le E_P$, where, writing $d_i = \max(h_i - \mu_i,\ \mu_i - \ell_i)$ for the largest possible y-deviation in bucket i,*

$$E_P = \sum_i c_i\,|\mu_i - \hat{Y}|\,\frac{w_i}{2} + \sum_i c_i\,w_i\,d_i.$$

Proof. Start from the exact between part of S_{xy} and replace the unknown $\bar{x}_i - \bar{X}$ by computable pieces plus error. Using $\bar{x}_i = m_i + \delta_i$ and $\bar{X} = \hat{X} + \Delta$ (Lemma A),

$$\bar{x}_i - \bar{X} = (m_i - \hat{X}) + (\delta_i - \Delta).$$

Also $\bar{Y} = \hat{Y}$ exactly (Lemma A), so $\mu_i - \bar{Y} = \mu_i - \hat{Y}$. Substitute into the between sum:

$$\sum_i c_i(\bar{x}_i - \bar{X})(\mu_i - \bar{Y}) = \sum_i c_i\big[(m_i - \hat{X}) + (\delta_i - \Delta)\big](\mu_i - \hat{Y}).$$

Distribute into three groups:

$$= \underbrace{\sum_i c_i(m_i - \hat{X})(\mu_i - \hat{Y})}_{=P_0} + \sum_i c_i\,\delta_i(\mu_i - \hat{Y}) - \Delta\sum_i c_i(\mu_i - \hat{Y}).$$

The last group vanishes. Why: Δ does not depend on i, so pull it out, and

$$\sum_i c_i(\mu_i - \hat{Y}) = \Big(\sum_i c_i\mu_i\Big) - \hat{Y}\sum_i c_i = N\hat{Y} - \hat{Y}\,N = 0,$$

using the definition $\hat{Y} = \frac{1}{N}\sum_i c_i\mu_i$. So the entire Δ term is $-\Delta \cdot 0 = 0$. This is the key cancellation: the error in our x-mean stand-in does *not* corrupt the between part of S_{xy} at first order.

What remains, after adding back the within part $\sum_i B_i$:

$$S_{xy} - P_0 = \underbrace{\sum_i c_i\,\delta_i(\mu_i - \hat{Y})}_{\text{(I): unknown } x\text{-placement}} \quad + \quad \underbrace{\sum_i B_i}_{\text{(II): within-bucket spread}} \quad .$$

Bound each piece using only what the summaries give us.

Piece (I). The only unknown is δ_i, and $(\star)$ says $|\delta_i| \le w_i/2$. Everything else is computable. So

$$\Big|\sum_i c_i\,\delta_i(\mu_i - \hat{Y})\Big| \le \sum_i c_i\,|\mu_i - \hat{Y}|\,|\delta_i| \le \sum_i c_i\,|\mu_i - \hat{Y}|\,\frac{w_i}{2}.$$

That is exactly the first sum in E_P.

Piece (II). Inside bucket i, bound the product term by term. Both x_{ij} and $\bar{x}_i$ lie in the span $[a_i, b_i]$, so their difference is at most the width: $|x_{ij} - \bar{x}_i| \le w_i$. And every y_{ij} lies in $[\ell_i, h_i]$ while $\bar{y}_i = \mu_i$, so $|y_{ij} - \mu_i| \le \max(h_i - \mu_i, \mu_i - \ell_i) = d_i$. Therefore

$$|B_i| \le \sum_j |x_{ij} - \bar{x}_i|\,|y_{ij} - \mu_i| \le \sum_j w_i\,d_i = c_i\,w_i\,d_i,$$

and summing over buckets gives the second sum in E_P.

Adding the two bounds via the triangle inequality yields $|S_{xy} - P_0| \le E_P$. □

Plain meaning: P_0 is our computable estimate of S_{xy}. It is wrong by at most E_P, and E_P has two clearly identified causes – not knowing the horizontal placement (piece I) and not knowing how points scatter inside a bucket (piece II). Both are controlled purely by widths, counts, and the stored min/max.

6 Step 4 — bounding S_{xx} (Lemma C)

Now the denominator. Define the computable estimate

$$Q_0 = \sum_i c_i\,u_i^2, \qquad u_i := m_i - \hat{X}.$$

Lemma 3. *$Q_0 - E_\downarrow \;\le\; S_{xx} \;\le\; Q_0 + E_\downarrow + E_\uparrow$, where*

$$E_\downarrow = \sum_i c_i\,|m_i - \hat{X}|\,w_i, \qquad E_\uparrow = \sum_i c_i\Big(\tfrac{w_i}{2} + D\Big)^2 + \sum_i c_i\,\tfrac{w_i^2}{4}.$$

Proof. Take the exact between part of S_{xx} and substitute $\bar{x}_i - \bar{X} = u_i + e_i$ where $e_i := \delta_i - \Delta$ collects all the x-placement uncertainty. Expanding the square:

$$\sum_i c_i(\bar{x}_i - \bar{X})^2 = \sum_i c_i u_i^2 \;+\; 2\sum_i c_i u_i e_i \;+\; \sum_i c_i e_i^2.$$

The first sum is Q_0. In the cross sum, replace e_i by $\delta_i - \Delta$; the $-\Delta$ piece drops out because $\sum_i c_i u_i = \sum_i c_i(m_i - \hat{X}) = N\hat{X} - \hat{X}N = 0$ (same kind of cancellation as before). So $\sum_i c_i u_i e_i = \sum_i c_i u_i \delta_i$. Putting the within part $\sum_i A_i$ back:

$$S_{xx} = Q_0 \;+\; 2\sum_i c_i u_i \delta_i \;+\; \underbrace{\sum_i c_i e_i^2}_{\ge 0} \;+\; \underbrace{\sum_i A_i}_{\ge 0}. \qquad (\dagger)$$

The last two sums are sums of squares, hence ≥ 0. The middle (signed) term is controlled by $(\star)$:

$$\left|2\sum_i c_i u_i \delta_i\right| \leq 2\sum_i c_i\,|u_i|\,|\delta_i| \leq 2\sum_i c_i\,|m_i - \hat{X}|\,\frac{w_i}{2} = \sum_i c_i\,|m_i - \hat{X}|\,w_i = E_{\downarrow}.$$

Lower bound. In (†), the two squared sums only add to S_{xx}, so dropping them can only decrease the right-hand side:

$$S_{xx} \geq Q_0 + 2\sum_i c_i u_i \delta_i \geq Q_0 - E_{\downarrow}.$$

Upper bound. Bound the two non-negative sums from above.

For $\sum_i c_i e_i^2$: $e_i = \delta_i - \Delta$, and $|\delta_i| \leq w_i/2$ by $(\star)$ while $|\Delta| \leq D$ by Lemma A, so $|e_i| \leq w_i/2 + D$ and $e_i^2 \leq (w_i/2 + D)^2$. Hence $\sum_i c_i e_i^2 \leq \sum_i c_i (w_i/2 + D)^2$, the first sum in $E_{\uparrow}$.

For $\sum_i A_i$: $A_i = \sum_j (x_{ij} - \bar{x}_i)^2$ is the total squared spread of c_i points confined to an interval of width w_i. That spread is largest when the points are pushed as far apart as the interval allows, i.e. split between the two endpoints; a short calculation shows the maximum is $c_i w_i^2/4$. So $A_i \leq c_i w_i^2/4$ and $\sum_i A_i \leq \sum_i c_i w_i^2/4$, the second sum in $E_{\uparrow}$.

Substituting both into (†) and using $|2\sum c_i u_i \delta_i| \leq E_{\downarrow}$:

$$S_{xx} \leq Q_0 + E_{\downarrow} + E_{\uparrow}. \qquad \square$$

Plain meaning: Q_0 is our computable estimate of S_{xx}. Unlike S_{xy}, the uncertainty here is one-sided in character – the unknown placement and the within-bucket spread can only *inflate* S_{xx} relative to the estimate (that is why the upper error $E_{\uparrow}$ is separate and larger), while the lower side is controlled by the single signed term $E_{\downarrow}$.

7 Step 5 — from the two boxes to a slope interval

We now know, with certainty,

$$S_{xy} \in [\,P^-, P^+\,], \quad P^{\pm} = P_0 \pm E_P, \qquad S_{xx} \in [\,Q^-, Q^+\,], \quad \begin{array}{l} Q^- = Q_0 - E_{\downarrow}, \\ Q^+ = Q_0 + E_{\downarrow} + E_{\uparrow}. \end{array}$$

So the true pair (S_{xy}, S_{xx}) lies somewhere in the rectangle $[P^-, P^+] \times [Q^-, Q^+]$. We want the range of the ratio $\beta = S_{xy}/S_{xx}$ over that rectangle.

Theorem 1. *Suppose $Q^- > 0$ (the routine checks this; otherwise see the remark below). Let*

$$C = \left\{ \tfrac{P^-}{Q^-},\ \tfrac{P^-}{Q^+},\ \tfrac{P^+}{Q^-},\ \tfrac{P^+}{Q^+} \right\}$$

be the four ratios at the rectangle's corners. Then

$$\min C \ \leq\ \beta\ \leq\ \max C.$$

Proof. Consider $f(p,q) = p/q$ on the rectangle. Because $Q^- > 0$, the whole rectangle has $q > 0$, so f is finite throughout. Hold q fixed: f is linear in p, hence monotone in p. Hold p fixed: $\partial f/\partial q = -p/q^2$ has a constant sign on the rectangle (the sign of $-p$, and p does not change sign because $[P^-, P^+]$ is one interval), hence f is monotone in q. A function that is monotone in each coordinate separately on a rectangle attains its maximum and minimum at the rectangle's corners.

So the extreme values of f over the rectangle are among the four corner ratios in C. Since the true (S_{xy}, S_{xx}) is some point of the rectangle, its ratio β is between the smallest and largest of those corner ratios. $\square$

The degenerate case. If $Q^- \le 0$ (or the estimate $Q_0 \le 0$), the denominator interval $[Q^-, Q^+]$ straddles 0. Then β is genuinely unbounded – some point clouds consistent with the summaries make S_{xx} arbitrarily small and the slope blows up – so the only honest answer is $(-\infty, +\infty)$, and that is what the routine returns. It is a valid (if useless) enclosure.

8 What the bound is, in one paragraph

We could not compute the slope because the raw points were discarded and many point clouds fit the same summaries. We replaced the two unknown means by computable midpoint surrogates (Lemma A), split each of S_{xx}, S_{xy} into an exact between/within decomposition, and bounded every place an unknown entered using only two facts: each point's x sits within its bucket's width w_i of the centroid $(\star)$, and each point's y sits within the stored $[\ell_i, h_i]$. That produced a computable estimate and a computable error bar for the numerator $(P_0 \pm E_P)$ and for the denominator (Q_0, with $-E_\downarrow$ / $+E_\downarrow + E_\uparrow$). The true (S_{xy}, S_{xx}) lives in the resulting rectangle, and a ratio is monotone on a positive rectangle, so the four corner ratios bracket the true slope. Every error term is proportional to a bucket *width*: shrink the buckets and the interval shrinks with them. That is the whole proof, and the practical lever.

Numerical note. Everything above is exact in real arithmetic. A floating-point implementation should widen the final interval by the accumulated round-off (roughly machine epsilon times the magnitude of the sums). That is slack, not a gap in the argument.